UnboxAI System Specification

ArguAgent + UnboxAI

1. First screen asks for Participant Number (Ask the Researcher)
2. English or Hebrew

Details for the AI that the human fill:
3 screens:

1. *a position you would enjoy working in one year from now,*
2. *five things about yourself that would make you a good candidate for that role*
3. *one sentence about relevant experience you have. Please do not include your name or any employer names.*

The AI should first deliver a half-baked.

We use a custom-built text-based conversational AI interface developed at milab, powered by Google Gemini. The interface presents a chat window and, depending on condition, a Response Controls panel containing three sliders (1-5 scale):

- **Verbosity** (פירוט): How much the LLM elaborates - from very concise (1) to expansive (5). Baseline: 2.
- **Certainty** (ביטחון): How confidently the text asserts the candidate's case - from cautious/hedging (1) to assertive (5). Baseline: 4.
- **Self-promotion intensity** (הבלטה עצמית): How emphatically the text presents the participant's achievements - from modest/understated (1) to bold/emphatic (5). Baseline: 4.

The baseline values are identical across all conditions and deliberately offset from the modal self-presentation style of the student population on the two most personally loaded dimensions (certainty, self-promotion). The rationale follows directly from psychological ownership: control over the AI's tone can only generate ownership if there is a gap between the AI's default tone and the participant's own - if the default already matched how the participant would write, there would be nothing to adjust and the control manipulation (C3) would be inert. An off-style baseline gives C3 participants something meaningful to bring back toward themselves, and gives C2 participants a visible setting they can recognize as not-quite-theirs.

- **C1 - Hidden:** The Response Controls panel is not displayed. Sliders are applied server-side at baseline without participant awareness.

- **C2 - Visible, Locked:** The panel is displayed with sliders at baseline, greyed out with a lock indicator. Participants can see the positions and values (1–5) but cannot adjust them.
- **C3 - Visible, Controllable:** The panel is displayed and fully interactive from the moment the first draft appears. All adjustments are logged (slider, value, timestamp, turn number).

In all conditions, the AI does not disclose its parameter settings in conversation, and participants cannot request the chat to change its tone. Chat instructions about the letter's content (facts, structure, emphasis of specific experiences) are honored in all conditions; the AI's stylistic tone remains governed by the slider parameters.

In addition, **behavioral and process logs** will be collected server-side:

- number of prompts/messages sent,
- number of slider changes (C3 only),
- number of prompts sent after the "Ready to Submit" button appears,
- total characters,
- timestamps,
- submit-click time,
- and the final version of the cover letter.

Research Plan

The study takes place in a quiet room in the media innovation lab (milab), with a table, chairs, a computer with the AI system installed, and a central recording device. Each session follows the sequence below.

Phase 1 - Onboarding. The experimenter reads the following instructions: *"Hi, my name is [name], and I'm a researcher at milab. Today you'll create a cover letter for a desired job together with an AI assistant. We're studying how people create application documents with AI assistance. Before we begin, please review and sign the consent form. The session will be screen-recorded and your data will be fully anonymized."* The participant signs the consent form (Appendix B).

Phase 2 - The Task. The researcher explains: *"To write your cover letter, the AI needs a few details. Please fill in: a position you would enjoy working in one year from now; five things about yourself that would make you a good candidate for that role; and one sentence about relevant experience you have. Please do not include your name or any employer names. The AI will write a first draft of your cover letter based on what you entered. You'll have time to improve it through conversation. The system keeps an eye on the letter as you work, and a 'Ready to Submit' button will appear once the AI judges the letter is good enough to submit. When you see the button, you can submit, but you can still continue editing until a 10-minute time-limit is reached."*

(Note: In reality, the "Ready to Submit" button is governed by a fixed timing rule 0 it becomes available after 5 minutes have passed - not by a quality judgment. This is a minor deception necessary for the study; see Deception section and Appendix H.)

For conditions with a visible panel, the experimenter adds the condition-specific instruction:

- **C2:** *"On the side you can see the AI's response controls, the settings that shape how it writes. They are fixed for this session and cannot be changed."*
- **C3:** *"On the side you have the AI's response controls, the sliders that shape how it writes. You can adjust them at any time as you work."*

The AI generates a deliberately rough first draft at baseline settings, so the participant is motivated to keep working on it. The participant then refines the letter via chat (and, in C3, via the sliders). A 10-minute limit runs in the background but is not shown to the participant; only a single "2 minutes remaining" notice appears near the end, to avoid sustained time pressure. The "Ready to Submit" button becomes available after 5 minutes, appearing only once the AI has finished generating its current response, so it does not look like a direct reaction to the participant's latest edit. The participant decides if and when to click it. At the 10-minute cap, the current letter is recorded as final. The experimenter is present but silent throughout.

Phase 3 - Post-Test and Interview. Participants complete the Psychological Ownership, Perceived Authorship, Perceived Authenticity, and Perceived Control questionnaires (Appendices C-F). A semi-structured interview is then conducted. Before this section begins, the researcher will inform the participant that it is going to be recorded by audio (Appendix G, Interview Script).

End of Session - Debrief. The researcher thanks the participant and conducts a debriefing session, explaining the true nature of the study (Appendix H). Participants are offered to take the cover letter home with them.